

CSA4001-MANAGEMENT INFORMATION SYSTEM FOR DATA OPTIMIZATION

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10) Draw a use case diagram for Library Management System is an automation system used to manage a library and the different resource management required in it like cataloguing of books, allowing check out and return of books, invoicing, user management, etc. The user can search for books details using few book properties (Book ID, Title, Author, and Publisher). Searching should return details about all the book copies that match the search query.

AIM

The aim of creating the use case diagram for a Library Management System is to model the interactions between various actors (like User, Librarian, and System) and the system's key functionalities, including book management, search operations, and user management.

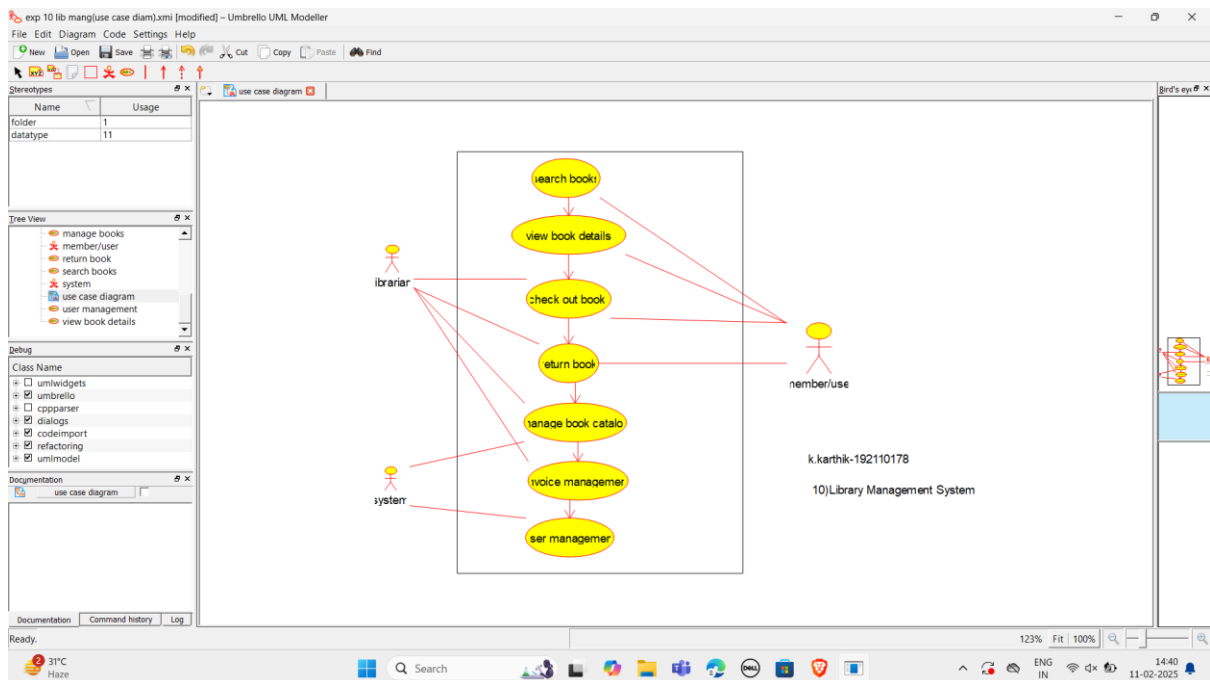
PROCEDURE:

1. Identify actors: User, Librarian, and System.
2. Define use cases: Search for books, Checkout book, Return book, Catalogue book, Manage user, Invoicing, Generate reports.
3. Establish relationships: Link actors to relevant use cases using association lines.
4. Include system boundary: Draw a box labelled Library Management System to demarcate the system's scope.
5. Use include/extend: Use include for mandatory actions like Search for books included in Checkout book; use extend for optional actions like Generate reports extending Invoicing.

6. Define relationships between use cases: Show use case interactions, such as Checkout book depending on Search for books and Manage user.
7. Finalize and review: Ensure completeness, accuracy, and usability of the diagram.

OUTPUT:

USE CASE DIAGRAM:



RESULT:

The Use Case Diagram will visually represent the actors interacting with the system, as well as the primary functionalities available to users and administrators of the library, ensuring that all core operations of book management, search, user management, and invoicing are clearly depicted.